Exploratory Data Analysis (EDA) Project Report

Project Overview

This project was completed as part of the CodeAlpha Data Analytics Internship. In this project, I performed Exploratory Data Analysis (EDA) on a customer churn dataset to understand customer behavior and identify the major factors that influence customer churn.

The main purpose of EDA is to explore the dataset, understand its structure, check data quality, identify patterns, and generate meaningful insights before applying advanced analytics or machine learning techniques.

During this project, I analyzed different aspects of customer data, including:

- Dataset structure and information
- Data quality and cleaning
- Customer churn behavior
- Customer demographics
- Contract types
- Monthly charges
- Customer tenure
- Internet service patterns
- Business insights

Dataset Source

For this analysis, I used the **Telco Customer Churn Dataset** available on Kaggle.

Dataset Source:
https://www.kaggle.com/datasets/blastchar/telco-customer-churn

The dataset contains customer-related information such as personal details, subscribed services, account information, charges, and churn status.

This dataset helps in understanding why customers leave a service and what factors can improve customer retention.

Tools and Technologies Used

The following tools and libraries were used during this project:

- Python
- Pandas
- NumPy

- Matplotlib
- Seaborn
- Google Colab

Dataset Overview

The original dataset contains information about:

- **7043 customers**
- **21 features**

The dataset includes different types of customer information:

Customer Information

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents

Service Information

- Phone Service
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming Services

Account Information

- Contract Type
- Payment Method
- Monthly Charges
- Total Charges

Target Variable

- Churn

Data Cleaning and Quality Check

Before starting the analysis, I performed data quality checks to make sure the dataset was clean and ready for analysis.

The following checks were performed:

- Missing value analysis
- Duplicate record checking
- Data type verification
- Blank value detection

Findings:

- No duplicate records were found in the dataset.
- Initially, no missing values were detected using the `isnull()` function.
- During further checking, the `TotalCharges` column was found to contain 11 blank values.
- These blank values were converted into missing values and removed.

After cleaning, the final dataset contained:

- **7032 rows**
- **21 columns**

The cleaned dataset was then used for further analysis.

Exploratory Data Analysis

1. Customer Churn Analysis

First, I analyzed customer churn distribution to understand how many customers stayed with the company and how many customers left.

Results:

- Customers who stayed: **73.42%**
- Customers who churned: **26.58%**

Insight:

Around one-fourth of customers have left the service, which indicates that customer retention strategies are important for reducing churn.

2. Customer Demographics Analysis

I analyzed customer demographics such as gender and senior citizen status to understand the customer base.

Gender Distribution:

- Male customers: **50.47%**
- Female customers: **49.53%**

Insight:

The customer base is almost equally divided between male and female customers.

Senior Citizen Analysis:

- Non-senior citizens: **83.76%**
- Senior citizens: **16.24%**

Insight:

Most customers belong to the non-senior citizen category.

3. Contract Analysis

I analyzed different contract types and their relationship with customer churn.

Contract Distribution:

- Month-to-month: **55.11%**
- Two year: **23.96%**
- One year: **20.93%**

Churn Rate:

- Month-to-month customers: **42.71%**
- One year contract customers: **11.28%**
- Two year contract customers: **2.85%**

Insight:

Customers with month-to-month contracts are more likely to leave the company, while customers with long-term contracts show better retention.

Businesses can reduce churn by encouraging customers to choose longer-term plans through loyalty programs and offers.

4. Monthly Charges Analysis

Monthly charges were analyzed to understand whether pricing affects customer churn.

Findings:

- Average monthly charge: **64.80**
- Minimum monthly charge: **18.25**
- Maximum monthly charge: **118.75**

Insight:

Customers with higher monthly charges show a higher tendency to churn, which indicates that pricing and service value may influence customer decisions.

5. Tenure Analysis

Customer tenure represents how long a customer has been using the service.

Findings:

Average customer tenure:

- Overall customers: **32.42 months**

Average tenure based on churn:

- Existing customers: **37.65 months**
- Churned customers: **17.98 months**

Insight:

Customers who leave the company usually have a shorter relationship period with the company.

Improving customer onboarding and providing better support during the initial months can help reduce churn.

6. Internet Service Analysis

Customer distribution based on internet service type was also analyzed.

Internet Service Distribution:

- Fiber optic: **3096 customers**
- DSL: **2416 customers**
- No internet service: **1520 customers**

Insight:

Fiber optic customers showed comparatively higher churn behavior. This may indicate possible issues related to pricing, service quality, or customer expectations.

Key Business Insights

Based on the complete analysis, the following insights were identified:

- Approximately **26.58% customers have churned**.
- Month-to-month contract customers have the highest churn rate.
- Long-term contracts help improve customer retention.
- Customers with shorter tenure are more likely to leave.
- Higher monthly charges may increase the possibility of churn.
- Fiber optic users show higher churn compared to other internet service users.
- Improving customer experience during the early stages can help reduce customer loss.

Conclusion

In this project, I successfully performed Exploratory Data Analysis on the Telco Customer Churn dataset.

The analysis helped in understanding customer behavior and identifying important factors affecting churn, such as contract type, customer tenure, monthly charges, and internet service type.

The insights generated from this analysis can help businesses create better customer retention strategies and improve overall customer satisfaction.